

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

CHITTIBOINA HARSHITHA
192473031RA

CO3-AT1-Assignment

View Only

[← Back](#)

QUESTIONS

Q1Banking Transaction System –
Exception Handling

10 marks

**Q2**Hospital Patient Registration –
User-Defined Exception

10 marks

**Q3**E-Commerce Order Processing
– Built-in Exceptions

10 marks

**Q4**Online Food Delivery – Multiple
Threads and Execution Order

10 marks

**Q5**Manufacturing Plant –
Producer-Consumer Problem

10 marks



5/5 answered

Q4

Online Food Delivery – Multiple Threads and Execution Order

10

marks

An online food-delivery application uses three threads: Order Processing, Payment Processing, and Delivery Assignment.

Develop a Java program in which these activities are represented by separate threads. Analyze the problem if all three threads are started independently.

Modify the program so that:

- Order processing completes before payment processing.
- Payment processing completes before delivery assignment.
- The program uses appropriate Java thread-management techniques.
- The output clearly shows the required execution order.

Explain the difference between simply starting all threads and explicitly controlling their execution order.

Sample Test Cases:

Input / Scenario Expected Result Analysis Focus

Order=Pizza; Payment=UPI; Delivery=Agent A Order →
Payment → Delivery Normal execution

Order=Burger; Payment=Card; Delivery=Agent B Order
→ Payment → Delivery Concurrency / design
Payment=Failed Delivery assignment should not
proceed Exception / boundary

Your Code

```
1 import java.util.Scanner;
2 public class onlinedelivery
3 {
4     static class order extends Thread
5     {
6         public void run()
7         {
8             System.out.println("order proces
9         }
10    }
11    static class payment extends Thread
12    {
13        public void run()
14        {
15            System.out.println("payment proc
16        }
17    }
18    static class delivery extends Thread
19    {
20        public void run()
21        {
22            System.out.println("delivered");
23        }
24    }
25    public static void main(String[] args) {
26        Scanner sc = new Scanner(System.in);
27        String order = sc.nextLine();
28        String payment = sc.nextLine();
29        String agent = sc.nextLine();
30        order t1 = new order();
31        payment t2 = new payment();
32        delivery t3 = new delivery();
33        t1.start();
34        t1.join();
```

```
38         t2.start();
39         t2.join();
40         t3.start();
41         t3.join();
42         System.out.println("order:" +order);
43         System.out.println("payement:" +paym
44         System.out.println("delivery:" +agen
45
46     }
47 }
```

Input

```
pizza
upi
agent a
```

Output

```
order processing completed
payment processinf compleetd
delivered
order:pizza
payement:upi
delivery:agent a
```

✓ Submitted